

SERVICE MANUAL

Invivo Corporation
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Document TR0358S

Revision 01

GE 1.5T HD 4 Channel Breast Array

GE Catalog Part Number: M3335SB

Invivo Part Number: 800358

TABLE OF CONTENTS

INTRODUCTION	3
MISCELLANEOUS.....	4
Product Identification and Shipping List.....	4
Compatibility	4
1-3 Related Documentation [Project Leader]	4
Environmental Requirements.....	5
LANDMARK SETUP	6
Coil Installation.....	6
Reverse Polarity Configuration [Optional]	6
Installing the Coil.....	6
Installation Functional Checks	6
Periodic Quality Assurance Check [GE Provides QA Procedure].....	6
QA	7
TROUBLESHOOTING	8
RF Signal Channels Check.....	8
PIN Diodes Check.....	8
REPLACEMENTS	10
FRU	15
Field Replaceable Units	15
Other Replaceable Accessories.....	15

INTRODUCTION

The purpose of this document is to archive the content that is sent to GE for use in their electronic service documentation. Six modules are sent to GE:

- Miscellaneous
- Landmark Setup
- QA
- Troubleshooting
- Replacements
- FRU

The resulting electronic service document is not the property of Invivo, however, the content contributed by Invivo is archived in this document.

MISCELLANEOUS

Product Identification and Shipping List

This is a service manual for the GE 1.5T HD Open Breast Array.



SHIPPING LIST – TABLE 1-1

Description	GE Part #	Invivo #
Electrical Assembly, 1.5T HD Open Breast Array	5145567-2	1 ea
Sternum Pad	2293670-8	1 ea
Ramp Pad	2293670-7	1 ea
Phantom Positioner	2246360-9	2 ea
Phantom Assembly	46-317586G1	1 ea
Operator's Manual	5145567-3	1 ea

Compatibility

This coil is compatible with the following hardware configurations

Signa Excite 14.0 - 1.5T & Value 1.5T

1-3 Related Documentation [Project Leader]

- 1.5T HD Open Breast Array Operator's Manual, 5145567-3

Environmental Requirements

Storage

This equipment must be stored under the following conditions:

- Ambient temperature range of +10°C to +40°C
- Relative humidity range of 10% To 90%, non-condensation
- Atmospheric pressure range of 500 hPa TO 1060 hPa

To store the coil, a storage space of 42.4 cm/16.7 in (width) x 40.6 cm/16.0 in (depth) x 18.6 cm/7.3 in (height) is required.

Dimensions

Coil:	42.4 cm x 40.6 cm x 18.6 cm	(16.7 in x 16.0 in x 7.3 in)
Phantom:	11.1 cm Spherical	(4.38 in Spherical)
Phantom Positioner:	14.0 cm DIA x 1.27 cm	(5.50 in DIA x 0.5 in)

Weight

Coil:	6.80 kg	15.00 lb
Phantom:	0.67 kg	1.467 lb
Phantom Positioner:	0.14 kg	0.310 lb

LANDMARK SETUP

Coil Installation

Reverse Polarity Configuration [Optional]

Installing the Coil

There is one configuration available:

HD 4Ch Breast Array

Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to: Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.
2. Perform Section 3 – Coil Imaging Performance Verification.

Periodic Quality Assurance Check [GE Provides QA Procedure]

QA checks should be performed during installation and during planned maintenance. Perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Coil Imaging Performance Verification.

QA



To: Ishiguro san
From: Larry Hyler
Date: November 16, 2006
Subject: **QA Procedure**

Historically we have received the QA procedure from GE and then we put the QA procedure into the Operator's Manual. It does not make sense for us to receive the QA procedure and then send it right back to you for the service documentation. So please send us the QA procedure and also use the same procedure for the service documentation.

Please let us know if this is acceptable.

Thank you.

TROUBLESHOOTING

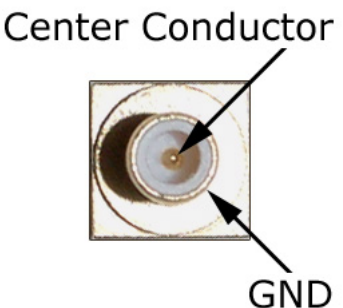
RF Signal Channels Check

NOTE: The center pins of the System Connector *receive coaxial connectors* are extremely fragile. **Do not attempt to make any measurements at these center pins.**

With the coil cable disconnected from the coil, use a multimeter to assure an open circuit condition exists between the designated pin and GND [coil side], for each signal channel.

EXTERNAL CABLE EXPECTED READINGS

Signal Channel	Coil Cable Connector	
MC-1	1	GND
MC-2	2	GND
MC-3	3	GND
MC-4	4	GND
GND	NOTE: The following points are common to GND: MC-1 through MC4 coaxial shields	

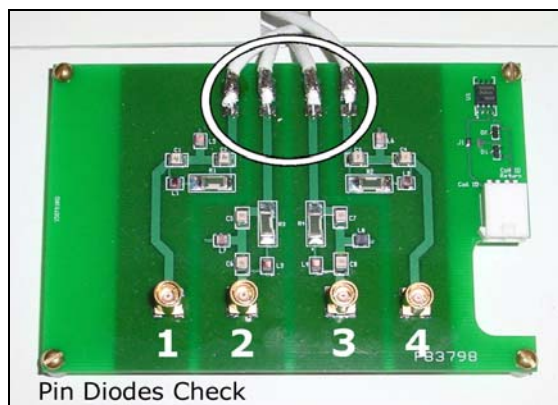


If one or more of the readings indicate a short, replace the cable. If all of the above readings indicate an open circuit condition, proceed to the **PIN Diodes Check**.

PIN Diodes Check

With the coil cable disconnected from the coil, use a multimeter to measure the diode drop across each of the coil coaxial cables.

Make the measurements at the cable solder connections as denoted by the oval on the right.



PIN DIODE EXPECTED READINGS

System Connector		Reading
Positive Lead	Negative Lead	
Center Conductor 1	GND [G1]	0.5 ± 0.1
GND [G1]	Center Conductor 1	Open
Center Conductor 2	GND [G2]	0.5 ± 0.1
GND [G2]	Center Conductor 2	Open
Center Conductor 3	GND [G3]	0.5 ± 0.1
GND [G3]	Center Conductor 3	Open
Center Conductor 4	GND [G4]	0.5 ± 0.1
GND [G4]	Center Conductor 4	Open

If one or more of the readings indicate a short, replace the coil. If all of the above readings are correct, redo *Coil Imaging Performance Verification*. If the problem still exists, replace the coil.

REPLACEMENTS

Replacements

1.5T 4 Channel HD Open Breast Coil Cable Replacement

([Click here for Summary Version](#))

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	30 mins	Timing Not Applicable

2 Overview

Follow this process to replace the cable assembly on the 1.5T 4 Channel HD Open Breast Coil by Invivo (M3335SB).

Last Update	XXXXX
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3 Preliminary Requirements

3.1 Tools and Test Equipment

- Philips screw driver
- Flat blade screw driver

3.2 Consumables

No consumables required.

3.3 Replacement Parts

5145567-5 Cable

3.4 Safety

No general safety information.

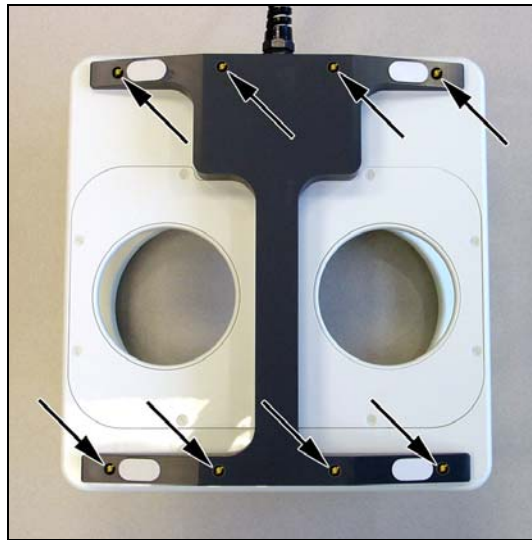
3.5 Required Conditions



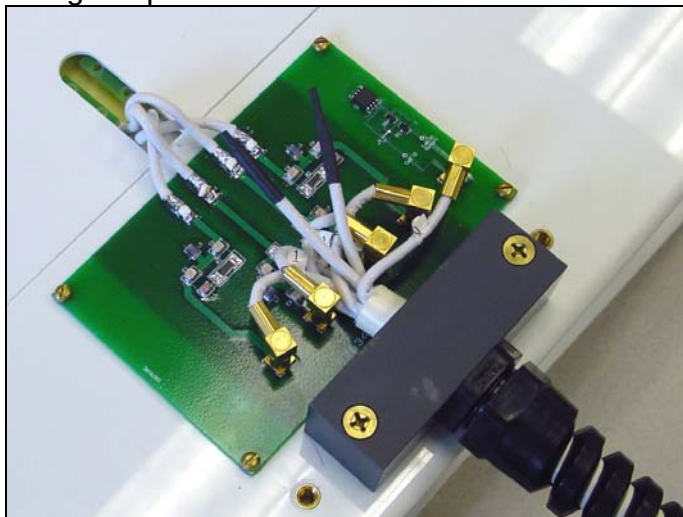
DURING CABLE REPLACEMENT, THE 1.5T 4 CHANNEL HD OPEN BREAST COIL PHASE SHIFTER BOARD IS EXPOSED. **USE ESD SAFETY PRECAUTIONS**

4 Procedure

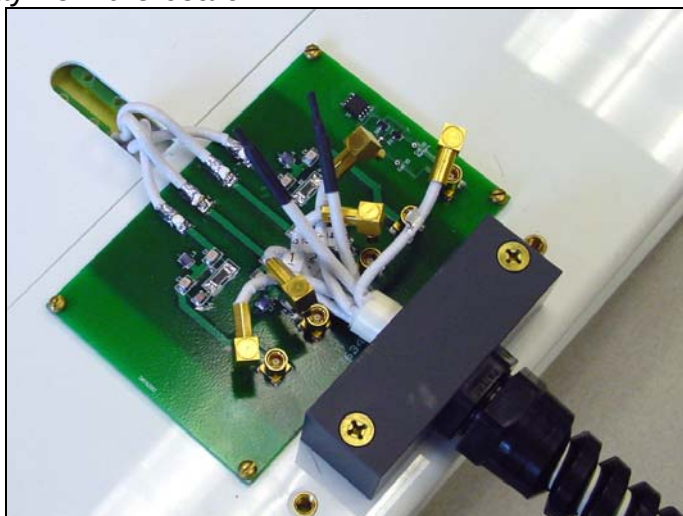
1. Remove the eight (8) brass slotted head screws from bottom of coil that hold the rib in place.



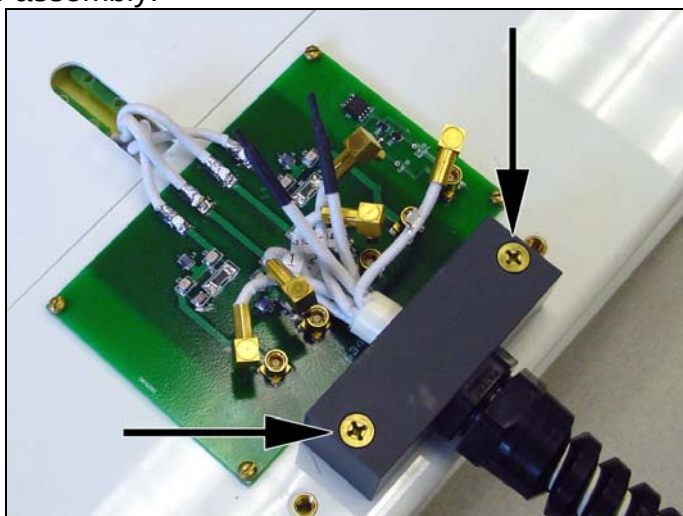
2. Remove the rib, revealing the phase shifter board.



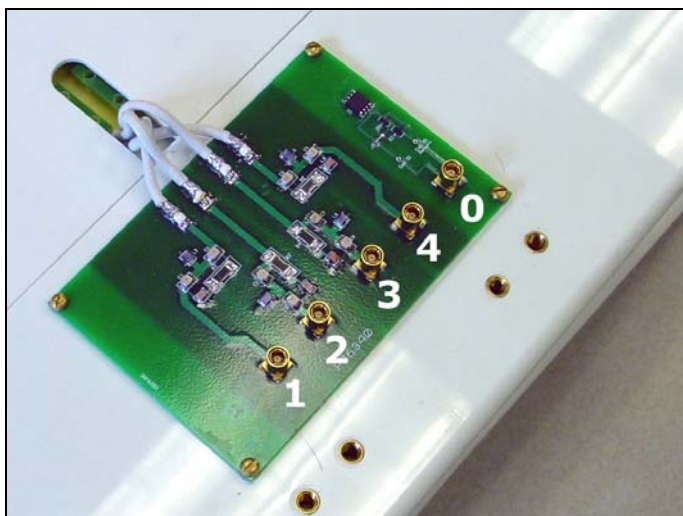
3. Unfasten the five (5) coaxial connectors from the phase shifter board. To unfasten, pull the connector away from the board.



4. Remove two (2) brass Phillips head screws that hold the cable assembly in place.
Remove the cable assembly.



5. Before installing the new cable assembly, make note of the cable numbering designations.



6. The numbered tabs on the five (5) coaxial cables correspond to the numbering designations above.



7. Reverse steps 1 through 4 to install the new cable assembly.

5 Finalization

Perform the Coil Imaging Performance Verification procedure.

FRU

Field Replaceable Units

FIELD REPLACEABLE UNITS LIST

Description	GE Part #	Invivo #
1.5T HD Open Breast Array with Cable	5145567-2	870251
Cable Assembly	5145567-5	870252
Phantom Positioner	2246360-9	870253

Other Replaceable Accessories

The following table lists replaceable accessories that may be purchased from Invivo Corporation.

OTHER REPLACEABLE ACCESSORIES LIST

Description	GE Part #	Invivo #
Ramp Pad	2246360-5	870254
Sternum Pad	2246360-6	870255